ANANTA COMPANIES

Information & Communication Technology

DocMan Alert

System Workflow & Architecture

End-to-end workflow, component communication, and how data moves through the system.

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Audience ICT, Compliance, and system administrators

Classification Internal

1. Purpose

DocMan Alert is Ananta Companies' internal tracker for factory compliance documents, employee licenses, and health checkups. A single Flask application computes expiry status in Bangladesh Standard Time, stores uploaded files, and emails the assigned person, department, or company when a record needs attention — one SQLite database, local file storage, one office SMTP relay.

2. System context

Users work in a browser on the office LAN; the application listens on port 5001, served by Waitress in production (run.py for development). Windows Task Scheduler starts the server and runs the background jobs.

Actor or system	What it does
Browser	Renders Jinja templates. Submits forms with a CSRF token. Downloads and views files.
DocMan Alert	Flask application. Enforces login, role, and factory scope. Writes SQLite and files.
SQLite database	instance/compliance.db. Companies, people, documents, alert history, and logs.
Upload folder	app/uploads by default. Versioned files for documents, licenses, and checkups.
SMTP relay	192.168.10.6 port 25. Sends mail as document_tracker@ananta.com.bd.
Tesseract OCR	Local engine. Reads scanned PDFs and images. No cloud OCR.
Task Scheduler	Starts the server, backs up daily, recalculates status daily, emails on Saturday.

Figure 1. Systems the tracker talks to.

3. Architecture

The application is one process — no separate frontend build, message queue, or microservices. A request enters Waitress, Flask routes it to a blueprint, SQLAlchemy reads or writes the database, and a Jinja template returns HTML; files stay on disk, never in the database.

Layer	Technology	Responsibility
Frontend	Jinja2, Bootstrap, small page scripts	Screens, forms, cascading dropdowns.
HTTP server	Waitress, port 5001, 12 threads	Production process. serve.py.
Application	Flask app factory, blueprints	Routes, validation, RBAC, status, mail.
ORM	Flask-SQLAlchemy, SQLite WAL	Tables, relationships, schema upgrades.
Files	Werkzeug uploads under UPLOAD_FOLDER	PDF, JPG, PNG, DOCX, XLSX up to 10 MB.
Mail	Flask-Mail over SMTP	One message per document per recipient.
OCR	PyMuPDF and Tesseract	Suggest fields from a PDF or image.

Figure 2. Layers inside one deployable application.

Request path

- Flask-Login, roles_required, and factory checks gate every request. The session cookie is not permanent and there is no idle timeout.
- The route loads data, applies no business logic the status engine or recipient resolver hasn't already defined, and renders a template.

- An after-request hook adds no-store caching and the standard security headers.

Nine blueprints split the routes without splitting the process: auth, main, documents, employees, admin, users, regulatory, alerts, and logs.

4. Frontend

Every screen is server-rendered HTML — templates/base.html plus page templates. There is no single-page application and no client-side data store; the browser keeps only the session cookie.

Screen	Who uses it	What the user does
Login	Everyone	Username and password. Locked accounts see a wait message.
Dashboard	Signed-in users	Counts by status and factory, plus critical items.
My Alerts	Signed-in users	Records that can email this user, and why.
Alert History	Signed-in users, factory scoped	Every sent or failed alert email.
Documents	Scoped by factory	Add, edit, renew, OCR extract, archive, delete, download, send alert.
Employee Compliance	Scoped by factory	Employees, licenses, checkups, OCR extract, Excel import.
Companies / Factories / Departments	Super Admin, Administration	Organisations, factories, and their standing alert recipients.
Users	Super Admin, Factory Admin	Accounts, roles, factory, email.
Regulatory Directory	Super Admin	Category, authority, and service lists.
Backups	Super Admin	Create and download a zip of the database and files.
Test Alert	Super Admin	Send a test message to the ICT addresses only.
System Logs	Super Admin	Operational events. Not a business audit trail.

Figure 3. Main screens.

A few dropdowns refresh by small inline fetch calls: factory selection reloads the department list (shared, not factory-owned); Category cascades to Regulatory Authority to Service on the document form; and a fourth call refreshes the alert-recipient checklist and group mailboxes.

5. Backend

create_app() loads config.py, opens SQLite with WAL and foreign keys, runs idempotent schema upgrades, and registers the nine blueprints, security headers, and error pages (403, 404, 413, unexpected).

Modules

Module	Role
app/routes/*.py	HTTP actions. Each function checks login, role, and factory.
app/forms.py	WTForms validation for login, documents, employees, users, and regulatory records.
app/models.py	Tables and relationships.
app/utils.py	BST dates, status engine, recipient resolution, email send, alert history.
app/ocr_service.py	Reads a PDF or image and suggests fields; verifies the source file stayed unchanged. Does not save a document.
app/import_utils.py	Validates Excel rows. Nothing is created until the user confirms.

Module	Role
app/backup_service.py	SQLite snapshot plus the upload folder, zipped.
app/schema_upgrade.py	Idempotent startup upgrades: columns, indexes, Transport-role and department seeding. Never wipes data.

Figure 4. Backend modules.

Business rules live in one place: compute_status() is the only source of status, and resolve_alert_recipients() is the only source of who gets an email. Routes only call them.

6. APIs

The application is form-driven. The JSON endpoints exist only to fill dropdowns; they require a logged-in session and are not a public API.

Method and path	Access	Returns
GET /documents/api/departments/<factory_id>	Any signed-in user	The shared department list — not filtered by factory_id.
GET /documents/api/alert-users	Signed-in user; factory_id checked against scope	Selectable recipients and group mailboxes for a factory/department.
GET /regulatory/api/authorities/<category_id>	Super Admin, Administration	Authorities in that category.
GET /regulatory/api/services/<authority_id>	Super Admin, Administration	Services under that authority.

Figure 5. JSON endpoints used by the pages.

Everything else is an HTML form post — login, save, extract, send alert, delete, import, backup, logout — each carrying a CSRF token. Downloads are GET routes that check factory scope first.

7. Database

SQLite stores operational data in instance/compliance.db with WAL journal mode, a busy timeout, and foreign keys — WAL lets a reader continue while a backup or status update writes.

Organisation and access

- companies, factories, departments — a factory belongs to one company; departments are a shared list (not owned by a factory), paired with a factory only where used.
- roles and users — one role per user. Factory-scoped users have a factory_id; Super Admin, Administration, and Compliance Viewer do not.
- company_alert_recipients and department_alert_recipients link users to an organisation, always via users.email.

Compliance records

- documents / document_versions / document_alert_recipients — the current record, its versioned files, and its assigned people.
- employees, employee_compliance, health_checkups — mirrored by employee_compliance_versions, health_checkup_versions, license_alert_recipients, and checkup_alert_recipients.
- employees.receive_alerts is off by default; an employee's personal email is never used otherwise.

Alerts, OCR, and logs

- alert_dispatches stops a repeat urgent email for the same record, status, and expiry date.
- alert_emails / alert_email_items are the send history — recipient, subject, result.

- ocr_results holds extracted text and suggested fields. system_logs holds operational events only.

Regulatory data is a chain — service_categories → regulatory_authorities → services — and a document may point at one of each.

8. Authentication and authorization

Login checks the password against a Werkzeug hash; five failures lock the account for 10 minutes (persisted on the user row, so it survives a restart). A success clears the lock and writes LOGIN_SUCCESS; logout (POST only) is logged the same way. The session cookie is HttpOnly, SameSite=Lax, Secure only over HTTPS, and not permanent — it ends when the browser closes, with no idle timeout. SECRET_KEY is required or the process refuses to start.

Role	Factory scope	Notable rights
Super Admin	All factories	Users, factories, regulatory directory, backups, test alert, system logs, permanent delete.
Administration	All factories	Companies, departments, documents. Can link a regulatory service. Cannot open logs, backups, or the test alert.
Factory Admin	Own factory	Users in that factory, departments, documents, permanent delete inside the factory.
Compliance Manager	Own factory	Create and edit records. Archive. Permanent delete inside the factory.
Compliance User	Own factory	Create and edit records. Cannot archive or permanently delete.
Compliance Viewer	All factories	Read only. Cannot change recipients, documents, or employees.
Transport Manager	Own factory	Same rights as Compliance Manager, for Transport.
Transport User	Own factory	Same rights as Compliance User, for Transport.

Figure 6. Roles.

A factory-scoped user who requests another factory's record, file, or delete page receives 403; an expiry date earlier than the issue date is always rejected. Super Admin and Factory Admin can permanently delete a user by typing its username — blocked for one's own account, the protected default Super Admin, the last remaining Super Admin, and (for a Factory Admin) users outside their factory. Deleting a user clears the reference on their past records rather than cascading.

9. Status engine

Dates are compared in Bangladesh Standard Time (UTC+6) by one function, used on save, on the daily job, and when the dashboard opens. Opening the dashboard never sends mail.

Days until expiry	Status	Email schedule
No expiry, or more than 90 days	Valid	No expiry email
61 to 90	Expiring in 90 Days	Saturday 09:00, one email for this record
31 to 60	Expiring in 60 Days	Saturday 09:00, one email for this record
8 to 30	Expiring in 30 Days	Saturday 09:00, one email for this record
1 to 7	Expiring in 7 Days	Daily 09:00 urgent email, once per expiry date
0 or earlier	Expired	Daily 09:00 urgent email, once per expiry date

Figure 7. Status bands and when mail is sent.

10. End-to-end workflow

Two diagrams trace a record from upload to alert history: first how a document becomes a stored record, then how an expiring record becomes a delivered, logged email.

Figure 8a. Document & record lifecycle.

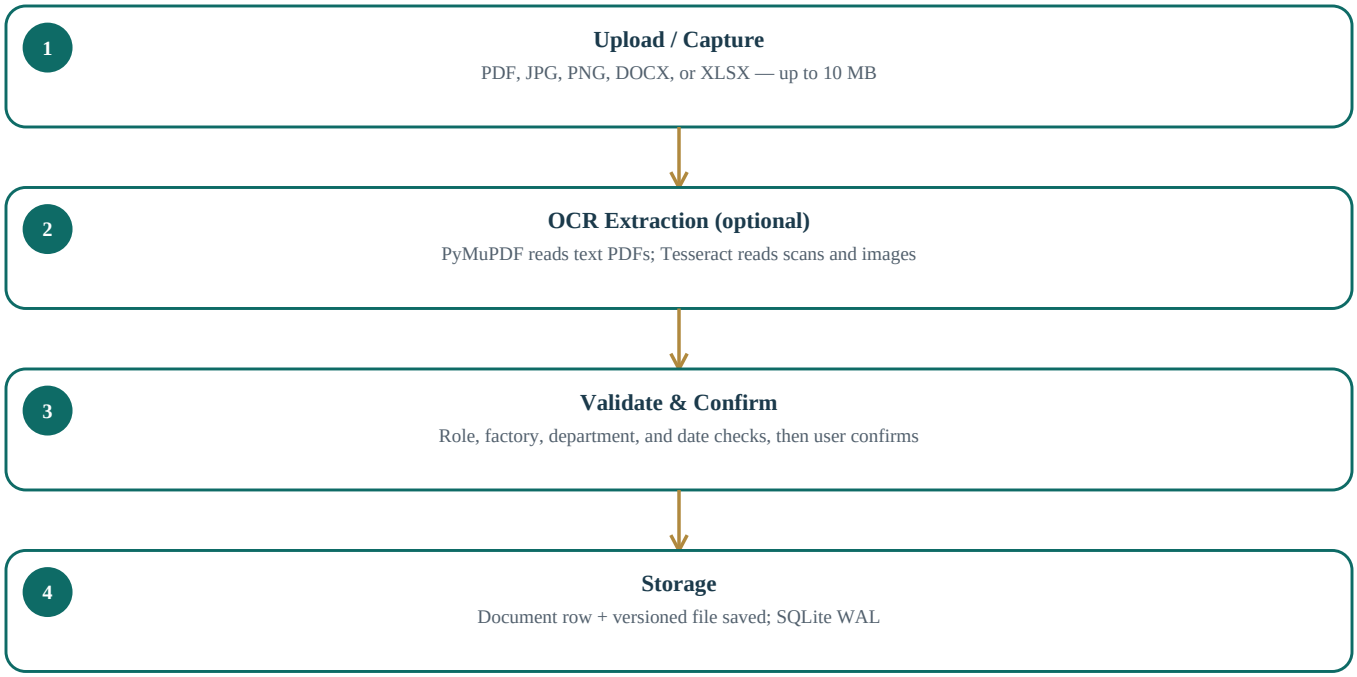
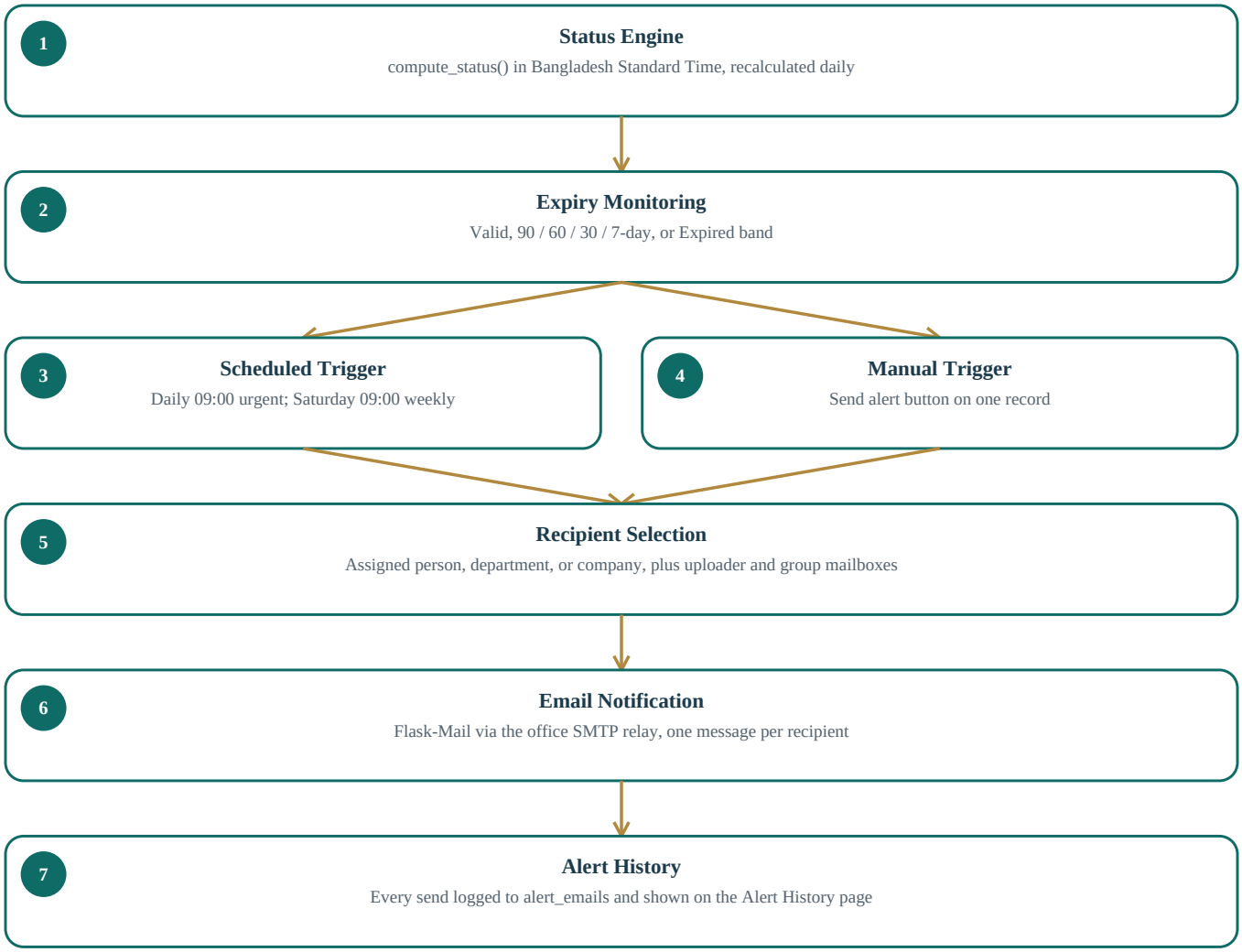


Figure 8b. Expiry monitoring & alert flow.



Section 11 gives the exact recipient-precedence rules and mailbox addresses behind step 4 above.

11. Notifications and alerts

There is no in-app inbox — notification means email, plus the Alert History page. Each message names one record: document, certificate number when present, company, factory, department, expiry, and status.

Who is addressed

- A specifically assigned person wins if present; otherwise the department's standing recipients; otherwise the company's — only one of these three applies.
- The uploader or creator is always included as well, independent of that chain.
- A factory-code mailbox is added when the factory matches a PRODUCTION_ALERT_ROUTES entry — today, Head Office only (arifkhan@ and badhon_admin@ananta.com.bd).
- A department-name mailbox is added when the record's department is Compliance or Transport, regardless of factory (compliance_dept@ / transport_agl@ananta.com.bd).
- An employee's personal email is included only when receive_alerts is on and the address is valid.
- Inactive users and invalid addresses are skipped; a factory-scoped user only sees their own factory's mail.

How a message is sent

- Flask-Mail sends from document_tracker@ananta.com.bd using SMTP_PASSWORD/MAIL_PASSWORD from the environment; one message per recipient, so one failure doesn't block the rest.

- alert_emails / alert_email_items record every attempt — type (weekly, emergency, manual, test), recipient, and result.
- Urgent sends are deduplicated per record, status, and expiry date; a renewal (new expiry date) can alert again.
- The Super Admin test alert goes only to TEST_ALERT_RECIPIENTS (arko_ict@ and omar_ict@ananta.com.bd by default) — never to production mailboxes.

Manual send (/documents/send-alert/<id>) uses the same recipient rules and template, but does not mark the urgent dedup table, so the 09:00 job can still fire later.

12. Scheduled jobs

Windows Task Scheduler runs four tasks, created by scripts/setup_task_scheduler.bat, all as SYSTEM.

Task	When	Program	Effect
ComplianceTracker_Server	At startup	run_server.bat → serve.py	Waitress on port 5001.
ComplianceTracker_StatusRecalc	Daily 09:00	run_status_recalc.bat → scheduled_status_recalc.py	Recalculate status; urgent emails for 7-day and expired records.
ComplianceTracker_EmailAlert	Saturday 09:00	run_email_alert.bat → scheduled_email_alert.py	One email per record in the 90/60/30-day bands.
ComplianceTracker_Backup	Daily 18:30	run_backup.bat → backup.py	Zip the database and uploads; drop copies past retention.

Figure 9. Scheduled tasks, Bangladesh Standard Time. Each .bat is a shim that calls the venv Python on the named script, from scripts/.

Each job writes JOB_START and JOB_SUCCESS/JOB_FAILURE to system_logs. scripts/ also holds one-off, hand-run tools — schema migrations, data fixes, the regulatory-data seeder, and restore.py — none of them scheduled.

13. File storage and backup

An upload is checked for extension and size, then saved as factory_<id>/doc<id>_v<version>_<filename>; the database stores the relative path, and a path that escapes the upload folder is refused.

Archive hides a document and keeps its files. Permanent delete requires typing the exact document name, then removes the document, its versions, recipient links, dedup rows, and files — but not its Alert History entries. Employees, licenses, and checkups have no equivalent delete, only archive.

Backup uses the SQLite backup API for a consistent snapshot plus every uploaded file, zipped. An optional BACKUP_MIRROR_FOLDER gets a second copy; BACKUP_RETENTION_DAYS defaults to 30.

14. External services

Service	Where it is used	What crosses the boundary
Office SMTP 192.168.10.6:25	Weekly, urgent, manual, and test alerts	HTML email from document_tracker@ananta.com.bd. No TLS on the default relay.
Tesseract OCR	Extract / Reprocess OCR	Image or scanned PDF in, suggestions out. Aborts if the source file changes mid-scan.
PyMuPDF	Text PDFs	Embedded text read locally — no Tesseract call.
Windows Task Scheduler	Server, backup, status, Saturday mail	Starts a local Python process. No outbound API.

Service	Where it is used	What crosses the boundary
Excel workbooks	Employee import	Parsed in memory and previewed; confirm writes the database. Nothing unknown is auto-created.

Figure 10. External and local integration points.

No cloud storage, no external identity provider, no third-party notification service. Identity is the users table; delivery is the office mail relay.

15. Security controls

- Session, role, and factory checked on every business page, action, and file.
- CSRF token on every form post; password hashes only; SMTP password and SECRET_KEY only from the environment.
- Account lockout after repeated failures. The session cookie is not permanent and has no idle timeout.
- Upload type/size limits; server-generated filenames; downloads cannot leave the upload folder.
- Permanent delete requires the exact record name and a role allowed to delete.
- Error pages return no stack traces. System Logs (Super Admin only) record operational events, not a before/after business audit.

16. Deployment layout

Path or setting	Meaning
serve.py / run.py	Production entry (Waitress, 0.0.0.0:5001) / development entry — not for production.
instance/compliance.db	SQLite database.
app/uploads, backups/	Default UPLOAD_FOLDER and BACKUP_FOLDER; both overridable. BACKUP_MIRROR_FOLDER is an optional second copy.
SECRET_KEY	Required. Signs the session. No default.
SMTP_PASSWORD or MAIL_PASSWORD	Required for mail. No default in source.
SMTP_HOST, SMTP_PORT	Default 192.168.10.6 and 25.
TESSERACT_CMD, TESSDATA_PREFIX	Local OCR. Languages default to eng+ben.
OCR_MAX_FILE_MB, OCR_MAX_PDF_PAGES, OCR_TIMEOUT_SECONDS	OCR limits. Default 10 MB, 50 pages, 120 seconds.
TEST_ALERT_RECIPIENTS	Super Admin test-alert addresses. Default arko_ict@ and omar_ict@ananta.com.bd.

Figure 11. Where the running system keeps its data and settings.

This picture matches the application as of 8 October 2026: one Flask process, server-rendered pages, SQLite, local files, office SMTP, local OCR, and four Windows scheduled tasks.